



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

FULL SYLLABUS

23-09-2021

JEE(Main + Advanced) : LEADER & ENTHUSIAST COURSE - SCORE (ADVANCED)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C	C	B	D	C	B
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	B,C,D	B,C,D	A,C	A,B,C	A,B,C	A,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	2.00	5.00	2.00	5.00	4.00	5.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B	A	B	B	A	A
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	B,C	A,B,C	A,B,C,D	A,B,D	A,B,C	A,B
SECTION-II	Q.	1	2	3	4	5	6
	A.	10.00	3.00	2.50	19.00	2.00	2.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	D	C	D	B	D	D
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,C,D	B,C	A,B,C,D	A,C	B,C,D	B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	0.44	365.00	0.50	0.38	16.66 or 16.67	0.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. **Ans (C)**

Let principal quantum number of initially excited energy level be n . Photon of maximum energy is emitted during transitions $n \rightarrow 1$.

$$\text{hence, } E_{\max} = (E_n - E_1) \dots (1)$$

Since, difference between energy of two consecutive levels decreases when n increases, therefore, photon of minimum energy is emitted during transition $n \rightarrow (n-1)$

$$\text{hence, } E_{\min} = E_n - E_{(n-1)} \dots (2)$$

$$\text{But energy of } n\text{th level is given by, } E_n = \frac{E_1}{n^2}$$

$$\therefore \text{ from equation (1), } \frac{E_1}{n^2} - E_1 = 525.224\text{eV} \dots (3)$$

$$\text{and from equation (3), } \frac{E_1}{n^2} - \frac{E_1}{(n-1)^2} = 1.224\text{eV} \dots (4)$$

solving equation (3) and (4), and $E_1 = -54.4 \text{ eV}$

and $n = 5$

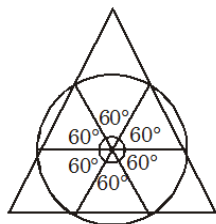
Hence, the gas is singly ionized Helium.

$\therefore$ (C)

2. **Ans (C)**

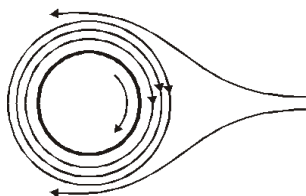
The area of cylinder inside prism is $\left(\frac{\pi}{3}R \times L\right) \times 3$

hence charge inside is $\frac{\pi}{3}R \times L \times 3 \times \sigma$



3. **Ans (B)**

As air rotates with the ball and also air comes from front relative velocity is less above and more below hence pressure is high above and low below causing the net force to be downwards.



4. **Ans (D)**

First glass receives heat thus it expands and then mercury expands more than glass.

5. **Ans (C)**

$$\text{Energy loss} = \frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} V_0^2$$

$$\Rightarrow \frac{1}{2} \frac{m}{2} V_0^2 = \frac{1}{4} m V_0^2$$

6. **Ans (B)**

Taking torque about pt of contact

$\Rightarrow$ torque on spool is clockwise, hence spool turns clockwise,

and goes right winding the string

SECTION-I (ii)

7. **Ans (B,C,D)**

When girls jump one by one total velocity of the system will be more. (because from conservation of momentum, velocity of car in this case will be

$$u_{\text{car}} = u \left(\frac{1}{M+2m} + \frac{1}{M+m} \right) m$$

$$\text{instead of } u \left(\frac{1}{M+2m} + \frac{1}{M+2m} \right)$$

when they jump simultaneously.

$\therefore$ (B, C, D)

9. **Ans (A,C)**

$$dV = -\vec{I} \cdot d\vec{r}$$

$$= - \left[(12xy^3 - 4x) \hat{i} + 18x^2y^2 \hat{j} \right] \cdot (dx \hat{i} + dy \hat{j})$$

$$= - [12xy^3 + 18x^2y^2 dy - 4x dx]$$

$$= - [d(6x^2y^3) - d(2x^2)]$$

It could be directly integrated without any reference to a path

$\Rightarrow$ field is conservative.

$$\text{Integrating, } V = 2x^2 - 6x^2y^3 + C$$

$$V = 0 \text{ at } (0, 0) \Rightarrow C = 0$$

$$V = 2x^3 - 6x^2y^3 \text{ at } (1m, 1m), V = -4V$$

$\therefore$ (a, c)

10. **Ans (A,B,C)**

$$R_{\text{eq}} = \frac{3 \times 6}{3 + 6} + 2 = 4\Omega$$

$$\text{Time constant} = \frac{L}{R_{\text{eq}}} = \frac{2}{4} = 0.5 \text{ sec}$$

in steady state, current through cell

$$= \frac{\varepsilon}{R_{\text{eq}}} = \frac{4}{4} = 1A$$

voltage across 2Ω resistor = $2V$

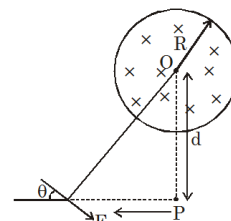
$\therefore$ (A, B, C)

11. **Ans (A,B,C)**

The friction acts backwards initially as point of contact has velocity in forward direction and when the cylinder stops, the friction pulls it back.

$\therefore$ (A, B, C)

12. **Ans (A,C,D)**



$$\int \vec{E} \cdot d\vec{\ell} = A \frac{dB}{dt}$$

$$E 2\pi \sqrt{x^2 + d^2} = \pi R^2 k$$

$$E = \frac{R^2 k}{2\sqrt{x^2 + d^2}}$$

$$W_{\text{ext}} = \int_0^\infty q\vec{E} \cdot d\vec{x} = \frac{q\pi R^2}{4} k$$

$\therefore$ (A), (C) and (D)

SECTION-II

1. **Ans (2.00)**

$$\text{Work done} = \int_{\vec{r}_1}^{\vec{r}_2} \vec{F} \cdot d\vec{r}$$

$$\text{then } W_1 = \int_0^a - (kx \hat{j}) \cdot dx \hat{i} = 0 \text{ and}$$

$$W_2 = \int_0^a -K (y \hat{i} + a \hat{j}) \cdot dy \hat{j} = -Ka \int_0^a dy = -ka^2$$

$$\therefore W = W_1 + W_2 = -ka^2$$

Hence n is -1, therefore n + 3 is 2.

Answer is 2.

2. **Ans (5.00)**

$$\frac{\ell_1 - \ell_2}{\ell_1 - \ell_3} = \frac{\alpha \ell_0 (T_1 - T_2)}{\alpha \ell_0 (T_1 - T_3)} = \frac{T_1 - T_2}{T_1 - T_3}$$

$$\text{or } \frac{5.2 - 3.9}{5.2 - 2.6} = \frac{80 - 60}{80 - T_3}$$

$$1.3 (80 - T_3) = 2.6 (20) \text{ or } T_3 = 40^\circ\text{C}$$

Answer is 5

3. **Ans (2.00)**

$$T = (273 + 27) \text{ K} = 300 \text{ K}$$

$$v = 1000 \text{ Hz}$$

$$\lambda = 0.2 \text{ m}$$

$$M = 124.5 \text{ gm/mol}$$

$$\text{Wave velocity } V = v\lambda = 200 \text{ m/s}$$

$$V = \sqrt{\frac{\gamma RT}{M}} \quad \gamma = 2$$

4. **Ans (5.00)**

$$\text{In first case } \lambda = \frac{v - v_s}{v} = \frac{340 - 30}{600} = \frac{310}{600}$$

$$\text{In second case } \lambda' = \frac{340}{600}$$

$$\text{So, } \lambda' - \lambda = \frac{340}{600} - \frac{310}{600} = \frac{30}{600} = 0.05 \text{ m}$$

$$\text{So, } \lambda' - \lambda = 5 \text{ cm.}$$

5. **Ans (4.00)**

Total increase in length of rods

$$= L\alpha\Delta T + \frac{L}{2}\alpha\Delta T = 3\frac{L}{2}\alpha\Delta T$$

Let the compression in spring A is x_A , B is x_B

$$\text{and C is } x_C \Rightarrow k_A x_A = k_B x_B = k_C x_C$$

$$\Rightarrow kx_A = 2kx_B = kx_C \Rightarrow x_A = 2x_B = x_C$$

$$\text{and } x_A + x_B + x_C = \frac{3}{2}L\alpha\Delta T$$

$$2x_B + x_B + 2x_B = 5x_B = \frac{3}{2}L\alpha\Delta T$$

$$\Rightarrow x_B = \frac{3}{10}L\alpha\Delta T$$

Energy stored

$$E = \frac{1}{2}kx_A^2 + \frac{1}{2}(2k)x_B^2 + \frac{1}{2}kx_C^2$$

$$= \frac{1}{2}k(2x_B)^2 + kx_B^2 + \frac{1}{2}k(2x_B)^2$$

$$= 2kx_B^2 + kx_B^2 + 2kx_B^2 = 5kx_B^2$$

$$E = 5k \left(\frac{9}{100}L^2\alpha^2\Delta T^2 \right)$$

$$= \frac{9}{20}k\alpha^2L^2\Delta T^2 = \frac{9}{5\beta}k\alpha^2L^2\Delta T^2 \Rightarrow \beta = 4$$

6. **Ans (5.00)**

$$\vec{r} = at^2 \hat{i} + bt^3 \hat{j} \text{ (m)}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = 2at \hat{i} + 3bt^2 \hat{j} \text{ (m/s)}$$

$$\vec{A} = 2a \hat{i} + 6bt \hat{j} = \frac{d^2\vec{r}}{dt^2} \text{ (m/s}^2\text{)}$$

At $t = 1 \text{ s}$

$$\vec{v} = 2a \hat{i} + 3b \hat{j} = 4 \hat{i} + 3 \hat{j} \text{ m/s}$$

$$\vec{A} = 2a \hat{i} + 6b \hat{j} = 4 \hat{i} + 6 \hat{j} \text{ m/s}^2$$

If radius of curvature is R then

$$\frac{|\vec{v} \times \vec{A}|}{|\vec{v}|} = \frac{|\vec{v}|^2}{R} \Rightarrow R = \frac{v^3}{|\vec{v} \times \vec{A}|} \quad v = 5 \text{ m/s}$$

$$\vec{v} \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 3 & 0 \\ 4 & 6 & 0 \end{vmatrix}$$

$$= \hat{i} (0 - 0) - \hat{j} (0 - 0) + \hat{k} (24 - 12) = 12 \hat{k}$$

$$\Rightarrow R = \frac{(5)^3}{12} \text{ m} = \frac{\eta^3}{12} \text{ m} \Rightarrow \eta = 5$$

PART-2 : CHEMISTRY

SECTION-I (i)

3. Ans (B)

When a chalk is dipped in ink, the surface retains the colour of ink due to adsorption of coloured molecules while solvent of ink goes deeper into stick due to absorption.

→ Precipitate of $Mg(OH)_2$ attains blue colour when precipitated in the presence of magneson reagent.

4. Ans (B)

$$d = 0.8 \text{ g/L}$$

$$r_{\text{gas}} = \sqrt{\frac{2}{3}} r_{O_2}$$

$$\text{or, } \frac{r_{\text{gas}}}{r_{O_2}} = \sqrt{\frac{32}{M}}; M = M_{\text{gas}}$$

$$\text{or, } \sqrt{\frac{2}{3}} = \sqrt{\frac{32}{M}} \Rightarrow M = 48 \text{ g}$$

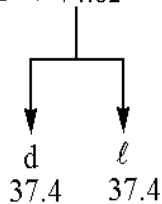
$$d = \frac{M}{V_m} \Rightarrow V_m = \frac{48}{0.8} = 60 \text{ L}$$

$$Z = \frac{P V_m}{RT} = \frac{1 \times 60}{0.08 \times 500} = \frac{6}{4} = 1.5$$

6. Ans (A)

$$\text{O.P.} \Rightarrow -\frac{3.5}{13.9} \times 100 \Rightarrow (-) 25.18\%$$

$$\text{Racemic mixture} \Rightarrow 100 - 25.18 \Rightarrow 74.82$$



$$\text{Total } d \Rightarrow 37.4$$

$$\text{Total } l \Rightarrow 37.4 + 25.18 = 62.6$$

SECTION-I (ii)

7. Ans (B,C)

(A) $Fe^{3+} \rightarrow 5$ unpaired electron

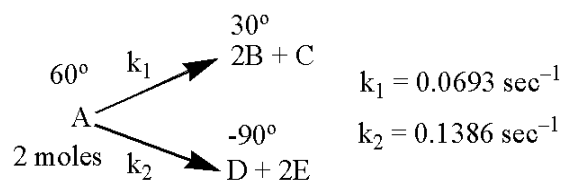
$Ni^{2+} \rightarrow 2$ unpaired electron

(B) $[Fe(CN)_6]^{3-}$ is more stable

(C) Across a period basic character ↓ and down the group, basic character ↑.

(D) I_2 is more polarisable.

9. Ans (A,B,C,D)



$$k = k_1 + k_2 = 0.0693 + 0.1386 = 0.2079$$

$$\theta_0 = 2 \times 60 = 120^\circ$$

$$\frac{C_B}{C_D} = \frac{2k_1}{k_2} = \frac{2 \times 0.0693}{0.1386} = \frac{0.1386}{0.1386} = 1$$

$$\therefore \text{At } t = \infty, [A] = 0$$

$$C_A = 2 - \frac{C_B}{2} - C_D = 0$$

$$2 = \frac{C_B}{2} - C_B \quad (\because C_B = C_D)$$

$$\text{or, } \frac{3}{2} C_B = 2, C_B = \frac{4}{3}$$

$$\therefore C_D = \frac{4}{3}$$

$$\theta_\infty = \frac{4}{3} \times 30 + \frac{4}{3} \times (-90)$$

$$= 40 - 120 = -80^\circ$$

$$t = \frac{1}{k} \ln \frac{\theta_0 - \theta_\infty}{(\theta_0 - \theta_t)}$$

$$\text{or, } \frac{10}{3} = \frac{1}{0.2079} \ln \frac{(120 + 80)}{(\theta_t + 80)}$$

$$\frac{2.079}{3} = \ln \frac{2000}{(\theta_t + 80)} \Rightarrow \frac{200}{\theta_t + 80} = 2$$

$$\theta_t = 20$$

After 75% completion :

$$C_A = 2(1 - 0.75) = 0.5$$

$$\therefore C_A = 2 - \frac{C_B}{2} - C_D = 0$$

$$\text{or, } 0.5 = 2 - \frac{C_B}{2} - C_B = 0$$

$$\text{or, } \frac{3C_B}{2} = 1.5 \Rightarrow C_B = 1; \therefore C_D = 1$$

$$\theta_t = 0.5 \times 60 + 60 + 1 \times 30 + 1 \times (-90)$$

$$= -30 \Rightarrow \text{leavorotatory}$$

$$(t_{1/2})_{\text{overall}} = \frac{0.693}{k} = \frac{0.693}{0.2079} = 3.33 \text{ sec} = \frac{10}{3}$$

% of C formed in the products

$$= \frac{k_1}{2k_1 + k_1 + k_2 + 2k_2} \times 100$$

$$\frac{k_1}{3(k_1 + k_2)} \times 100 = \frac{1}{3} \times \frac{0.0693}{0.2079} \times 100 = 11.11\%$$

10. Ans (A,B,D)

A → 0.3 M, 100 ml HCl + 50 ml water

$$[H^+] =$$

$$\frac{0.3 \times 100}{150} = \frac{30}{150} = \frac{1}{5} = 0.2 \times 2 \times 10^{-1}$$

$$pH = 1 - \log 2 = 1 - 0.3 = 0.7$$

B → 50 ml, 0.2 M HCl + 50ml 0.05 M NaOH

HCl = 10 mmole, NaOH = 2.5 mmole

$$H^+ = \frac{7.5}{100} = 7.5 \times 10^{-2} M$$

$$pH = 2 - \log 7.5 = 2 - 0.88 = 1.12$$

C → 50 ml, 0.2 M HCl + 100ml 0.05 M

NH₄OH



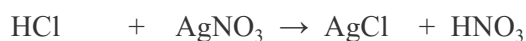
10 mmole 5 mmole

5 mmole 0 5 mmole

$$[H^+] = \frac{5}{150} = \frac{1}{30}$$

$$pH = -\log \frac{1}{30} = \log 30 = \log 3 + \log 10 = 1.48$$

D → 50 ml, 0.2 M HCl + 100ml 0.1 M AgNO₃



10 mmole 10 mmole

0 0 10 mmole 10 mmole

$$K_{sp} AgCl = [Ag^+][Cl^-]$$

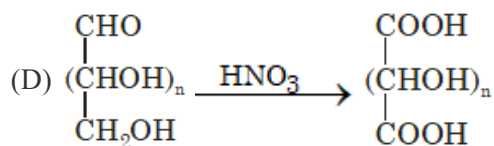
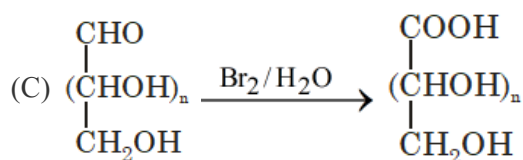
$$10^{-10} = [Ag^+]^2 \quad [\because [Ag^+] = [Cl^-]]$$

$$\text{or, } [Ag^+] = 10^{-5} M$$

$$\therefore \text{moles of } Ag^+ = 150 \times 10^{-5} \times 10^{-3} \text{ mole}$$

$$= 1.5 \times 10^{-6} \text{ mole}$$

12. Ans (A,B)



SECTION-II

3. Ans (2.50)

$$W = -1750 \text{ kJ}$$

$$W_{\text{irr}} = -1000 \text{ kJ}$$

$$\Delta S_{\text{total}} = \frac{q}{T} \text{ kJK}^{-1}$$

$$\Delta S = nR \ln \frac{V_2}{V_1} = \frac{nRT \ln V_2/V_1}{T} = \frac{-W_{\text{rev.}}}{T}$$

$$= \frac{1750}{300} \text{ kJK}^{-1}$$

$$\Delta S_{\text{surr.}} = \frac{-q}{T_{\text{surr.}}} = \frac{W_{\text{irr.}}}{T} = \frac{-1000}{300} \text{ kJK}^{-1}$$

$$\Delta S_{\text{total}} = \frac{1750}{300} - \frac{1000}{300} = \frac{750}{300} = 2.5 \text{ kJK}^{-1}$$

4. Ans (19.00)

Equimolar, $P_T = 45 \text{ torr}$

$$x_A \& x_B, P'_T = 22.5 \text{ torr} \frac{x_A}{x_B} = Y,$$

Initially, $x_A = x_B = 0.5$, $P_A^0 = 20 \text{ torr}$

$$P_T = P_A^0 x_A + P_B^0 x_B$$

$$\text{or, } 45 = 0.5 P_A^0 + 0.5 P_B^0$$

$$\text{or, } P_A^0 + P_B^0 = 90$$

$$\text{or, } P_B^0 = 90 - 20 = 70 \text{ torr.}$$

$$P'_T = (P_A^0 - P_B^0) x_A - P_B^0$$

$$\text{or, } 22.5 = (20 - 70) x_A + 70$$

$$\text{or, } 22.5 = -50 x_A + 70$$

$$\text{or, } 50 x_A = 47.5$$

$$\text{or, } x_A = \frac{47.5}{50} = 0.95$$

$$\therefore x_B = 0.05$$

$$\therefore Y = \frac{x_A}{x_B} = \frac{0.95}{0.05} = 19$$

5. Ans (2.00)

$$X = 4, Y = 2$$

X ⇒ Valine ($P^I = 5.97$), Arginine ($P^I = 10.76$),

Lysine ($P^I = 9.87$), Alanine ($P^I = 6.02$)

Y ⇒ Glutamic acid ($P^I = 3.22$), Aspartic acid

($P^I = 2.98$)

$$P^I = \frac{P^{K_{a1}} + P^{K_{a2}}}{2}$$

Amino acid having $P^I > pH$ moves towards cathode

Amino acid having $P^I < pH$ moves towards anode

6. Ans (2.00)

	Reagent	Compound-I	Compound-II
I	NaHCO ₃	×	✓
II	Na Metal	✓	✓
III	Neutral FeCl ₃	✓	×
IV	AgNO ₃ + NH ₄ OH	×	×
V	CS ₂ , HgCl ₂	✓	✓
VI	NaOI	×	×
VII	2,4-DNP	×	×

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (D)

$$\int_0^z \sin(f(x))(1-x) dx = (z^2 + z) \sin(f(z))$$

differentiate both sides w.r.t. 'z'

$$\sin f(z)(1-z) = (2z+1)\sin(f(z)) + (z^2+z)\cos(f(z))f'(z)$$

$$-3z\sin f(z) = (z^2+z)\cos(f(z))f'(z)$$

$$-3 = (z+1)\cot(f(z)).f'(z)$$

2. Ans (C)

$$[\vec{u} \quad \vec{v} \quad \vec{w}] = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix} [\vec{a} \quad \vec{b} \quad \vec{c}]$$

$$= [\vec{a} \quad \vec{b} \quad \vec{c}]^3$$

$$[\vec{a} \quad \vec{b} \quad \vec{c}] = \begin{vmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ 0 & \frac{\sqrt{3}}{2} & \frac{1}{2} \\ 0 & 2 & 3 \end{vmatrix} = \left(\sqrt{3} - \frac{1}{2} \right)$$

3. Ans (D)

Required circle is the circle described on O(0,0) & S(3,4) as diameter

$$(x-0)(x-3) + (y-0)(y-4) = 0$$

$$x^2 + y^2 - 3x - 4y = 0$$

4. Ans (B)

$$|z|^n = |z|^{n-2}(z^2 + z) + 1 \quad \dots(1)$$

from (1) $z^2 + z$ is real

$$z^2 + z = \bar{z}^2 + \bar{z}$$

$$\Rightarrow (z - \bar{z})(z + \bar{z} + 1) = 0$$

$$\text{Since } \operatorname{Re}(z) \neq \frac{1}{2}$$

$$\Rightarrow z = \bar{z} \Rightarrow z \text{ is purely real}$$

$$\text{from (1) } z^{n-1} + 1 = 0$$

$$\Rightarrow z = -1 \text{ is only solution}$$

5. Ans (D)

Degree of P(x) is 5 with leading coeff. one.

$\therefore$ degree of P'(x) is 4 with leading coeff. five.

Hence,

$$P'(x) = 5(x-1)(x-2)^2(x-3)$$

$$P'(4) = 5 \times 3 \times 4 \times 1 = 60$$

6. Ans (D)

$$R_3 \rightarrow R_3 - (\cos x) R_1$$

$$f(x) = \begin{vmatrix} \sec x & \cos x & \sec^2 x + \cot x \operatorname{cosec} x \\ \cos^2 x & \cos^2 x & \operatorname{cosec}^2 x \\ 0 & 0 & \cos^2 x - \sec x - \cot^2 x \end{vmatrix}$$

$$f(x) = (\cos^2 x - \sec x - \cot^2 x) (\cos x - \cos^3 x)$$

$$= \cos x \cdot \sin^2 x \cdot \left(\cos^2 x - \frac{1}{\cos x} - \frac{\cos^2 x}{\sin^2 x} \right)$$

$$= \cos x \cdot \sin^2 x \cdot \left(\frac{\cos^3 x \cdot \sin^2 x - \sin^2 x - \cos^3 x}{\cos x \sin^2 x} \right)$$

$$f(x) = \cos^3 x (\sin^2 x - 1) - \sin^2 x$$

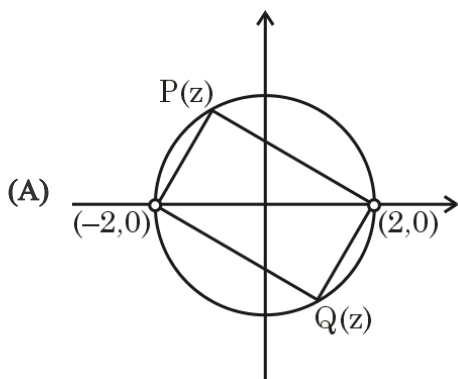
$$f(x) = -\cos^5 x - \sin^2 x$$

$$\int_{-\pi}^{\pi} -(\cos^5 x + \sin^2 x) dx = -2 \int_0^{\pi} \sin^2 x dx$$

$$= -4 \left(\frac{\pi}{4} \right) = -\pi$$

SECTION-I (ii)

7. Ans (A,C,D)

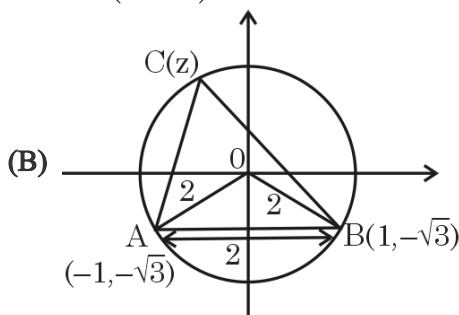


For P(z) we get

$$\arg\left(\frac{z-2}{z+2}\right) = \frac{\pi}{2}$$

For Q(z) we get

$$\arg\left(\frac{z-2}{z+2}\right) = -\frac{\pi}{2}$$



ΔAOB is equilateral Δ

$$\angle AOB = \frac{\pi}{3} \text{ and } \angle AOB = \frac{\pi}{6}$$

$$\arg\left(\frac{z-1+i\sqrt{3}}{z+1+i\sqrt{3}}\right) = \frac{\pi}{6}$$

$$\arg\left(\frac{z+1+i\sqrt{3}}{z-1+i\sqrt{3}}\right) = -\frac{\pi}{6}$$

$$\begin{aligned} \text{(C) } |z^2 - 1|^2 &= (z^2 - 1)(\bar{z}^2 - 1) \\ &= z^2 \bar{z}^2 - z^2 - \bar{z}^2 + 1 \end{aligned}$$

$$|z|^4 - 2(x^2 - y^2) + 1$$

$$= 17 - 2(x^2 - y^2) = 25 - 4x^2$$

$$(\because |z| = 2 \Rightarrow x^2 + y^2 = 4 \text{ where } x \in [-2, 2])$$

$$9 \leq |z^2 - 1|^2 \leq 25$$

$$\text{Hence } 3 \leq |z^2 - 1| \leq 5$$

8. Ans (B,C)

$$\sum_{i=0}^{100} \sum_{j=0}^{100} (C_i^2 + C_j^2 + C_i C_j) = a^{2n} C_n + 2^b$$

$$\sum_{i=0}^{100} (101 C_i^2 + {}^{200}C_{100} + C_i 2^{100})$$

$$101 {}^{200}C_{100} + 101 {}^{200}C_{100} + 2^{100} \cdot 2^{100}$$

$$202 {}^{200}C_{100} + 2^{200} = a^{2n} C_n + 2^b$$

$$a = 202; n = 100; b = 200$$

9. Ans (A,B,C,D)

$$g(x) = x + 2$$

$$f(g(x)) = 2 \tan^{-1}(x+2)$$

$$f^2(g(x)) - 5f(g(x)) + 4 > 0$$

$$f(g(x)) < 1 \text{ or } f(g(x)) > 4$$

$$\tan^{-1}(x+2) < \text{or } \tan^{-1}(x+2) > 2$$

$$x \in \left(-10, \tan\left(\frac{1}{2}\right) - 2\right)$$

$$\text{as } \frac{1}{2} < \frac{\pi}{6}$$

$$\Rightarrow \tan\left(\frac{1}{2}\right) < \frac{1}{\sqrt{3}}$$

$$\Rightarrow \tan\left(\frac{1}{2}\right) - 2 < \frac{1}{\sqrt{3}} - 2$$

Hence integers in range are $\{-9, -8, \dots, -2\}$

10. Ans (A,C)

$$P(\text{Prime number}) = \frac{1}{2}$$

$$\therefore p = \frac{1}{2}; q = \frac{1}{2}; n = 6$$

$$P(E) = P(r \geq 3)$$

$$= \frac{{}^6C_3 + {}^6C_4 + {}^6C_5 + {}^6C_6}{64}$$

$$= \frac{42}{64} = \frac{21}{32}$$

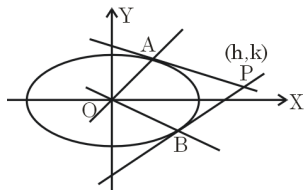
$$P(F) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$$

$$P(F/E) = \frac{P(F \cap E)}{P(E)} = \frac{P(F)}{P(E)}$$

$$= \frac{1}{8} \left(\frac{32}{21}\right) = \frac{4}{21} = \frac{a}{b}$$

$$\therefore a = 4 \text{ \& } b = 21$$

11. Ans (B,C,D)



Let the point of intersection of tangents at A and B be P(h, k) then equation of AB is

$$\frac{hx}{4} + \frac{ky}{1} = 1 \quad \dots(1)$$

Homogenising ellipse with (1) we get

$$\frac{x^2}{4} + \frac{y^2}{1} = \left(\frac{hx}{4} + \frac{ky}{1} \right)^2$$

$$\left(\frac{h^2 - 4}{16} \right) x^2 + (k^2 - 1) y^2 + \frac{2hk}{4} xy = 0 \quad \dots(2)$$

Combined equation of OA and OB is

$$x^2 + 4y^2 + \alpha xy = 0 \quad \dots(3)$$

Compare (2) and (3)

$$\frac{h^2 - 4}{16} = \frac{k^2 - 1}{4} = \frac{hk}{2\alpha}$$

$$\Rightarrow (h^2 - 4) = 4(k^2 - 1)$$

$$\Rightarrow h^2 - 4k^2 = 0 \Rightarrow x^2 - 4y^2 = 0$$

12. Ans (B,C)

$$[\vec{a} \ \vec{b} \ \vec{c}] = 0$$

So $\vec{a}$, $\vec{b}$, $\vec{c}$ are coplanar vectors

As $\vec{p}$ is equally inclined to $\vec{a}$, $\vec{b}$, $\vec{c}$.

$\Rightarrow \vec{p}$ is perpendicular to $\vec{a}$, $\vec{b}$, $\vec{c}$.

$$\therefore \vec{p} \cdot \vec{a} = \vec{p} \cdot \vec{b} = \vec{p} \cdot \vec{c} = 0$$

$$\therefore \vec{p} \cdot \vec{x} = \vec{p} \cdot \vec{y} = \vec{p} \cdot \vec{z} = 0$$

$$\vec{p} = \lambda (\vec{a} \times \vec{b}) = \lambda (-\hat{i} + 2\hat{j} + 3\hat{k})$$

$$\vec{p} \cdot (2\hat{i} - 3\hat{j} + 4\hat{k}) = \lambda(-2 - 6 + 12) \neq 0$$

SECTION-II

1. Ans (0.44)

$$f(x) = 3x^2 - 2ax + b \geq 0 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow (2a)^2 - 4(3)(b) \leq 0 \quad a^2 - 3b \leq 0$$

This is possible for 16 ordered pairs (a, b)

(1,1), (1,2), (1,3), (1,4), (1,5), (1,6), (2,2), (2,3), (2,4), (2,5), (2,6), (3,3), (3,4), (3,5), (3,6) & (4, 6)

$$\text{Hence probability} = \frac{16}{36} = \frac{4}{9} = 0.44$$

2. Ans (365.00)

$$S = a_{a_1} + a_{a_2} + \dots + a_{a_5}$$

$$a_1 = 13 - 2d; a_2 = 13 - d; a_3 = 13; a_4 = 13 + d;$$

$$a_5 = 13 + 2d$$

$$a_n = (13 - 2d) + (n - 1)d = (13 - 3d) + nd$$

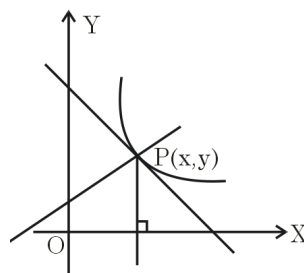
$$S = \sum_{n=1}^5 a_{a_n} = a_{13-2d} + a_{13-d} + a_{13} + a_{13+d} + a_{13+2d}$$

$$S = 5(13 + 10d)$$

Now $d \leq 6$ as terms are positive

$\therefore S_{\max}$ occurs at $d = 6$ and is equal to 365.

3. Ans (0.50)



$$\text{Equation of normal is } Y - y = \frac{-1}{m}(X - x)$$

$$X + mY - (x + my) = 0 \quad \dots(1)$$

perpendicular distance from (0, 0) to equation (1) is

$$\left| \frac{x + my}{\sqrt{1 + m^2}} \right| = |y|$$

$$\Rightarrow 2xy \frac{dy}{dx} = y^2 - x^2$$

$$\text{Put } y^2 = t \Rightarrow 2y \frac{dy}{dx} = \frac{dt}{dx}$$

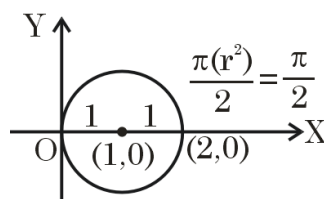
$$x \frac{dt}{dx} = t - x^2$$

$$\Rightarrow \frac{dt}{dx} - \frac{t}{x} = -x$$

$$\text{I.F.} = e^{-\int \frac{1}{x} dx} = \frac{1}{x}$$

$$\frac{t}{x} = -x + c \Rightarrow \frac{y^2}{x} = -x + c$$

(1,1) lies on it so $c = 2$

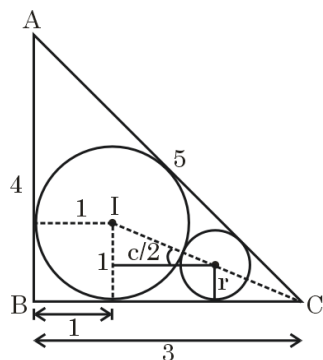


$$C : x^2 + y^2 - 2x = 0$$

$$\text{Area} = \frac{\pi}{2}$$

$$\therefore k = \frac{1}{2} = 0.50$$

4. Ans (0.38)



$$\text{Radius of first circle} = \frac{\Delta}{s} = 1$$

$$\sin \frac{C}{2} = \frac{1-r}{1+r} \quad \dots(1)$$

$$\text{Also } \sin C = \frac{4}{5} \Rightarrow \sin \frac{C}{2} = \frac{1}{\sqrt{5}}$$

$$\therefore r = \frac{\sqrt{5}-1}{\sqrt{5}+1} = \frac{1.236}{3.236}$$

$$r = 0.38$$

5. Ans (16.66 or 16.67)

for $r = 0$

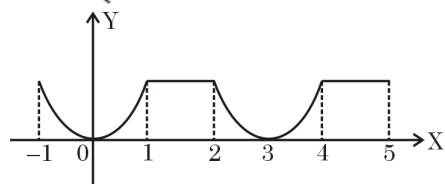
$$f(x) = \begin{cases} x^2 & ; -1 \leq x < 1 \\ 1 & ; 1 \leq x \leq 2 \end{cases}$$

for $r = 3$

$$f(x) = \begin{cases} (x-3)^2 & ; 2 < x < 4 \\ 1 & ; 4 \leq x \leq 5 \end{cases}$$

for $r = 6$

$$f(x) = \begin{cases} (x-6)^2 & ; 5 < x < 7 \\ 1 & ; 7 \leq x \leq 8 \end{cases}$$



$f(x)$ is periodic with period = 3

$$\int_0^{30} f(x) dx = 10 \int_0^3 f(x) dx$$

$$= 10 \left[\int_{-1}^1 x^2 dx + \int_1^2 1 \cdot dx \right]$$

$$= 10 \left(\frac{2}{3} + 1 \right) = \frac{50}{3}$$

$$= 16.66 \text{ or } 16.67$$

6. Ans (0.00)

$$a_{ij} = -a_{ji} \quad \forall i \& j$$

So A is a skew-symmetric matrix of order-3

$$\text{so } \det A = 0$$

$$\therefore \det(\text{adj}A) (\det(A))^2 = 0$$

JEE(Main + Advanced) : LEADER & ENTHUSIAST COURSE SCORE (ADVANCED)
ANSWER KEY
PAPER-2
PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,D	A,C	A,C	B,D	B,C,D	A,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	2.66 to 2.67	157.00 to 160.33	4.49 to 4.50	112.50	19.77 to 19.80	0.94 to 0.95
SECTION-III	Q.	1	2	3	4	5	6
	A.	5	4	2	5	5	8

PART-2 : CHEMISTRY

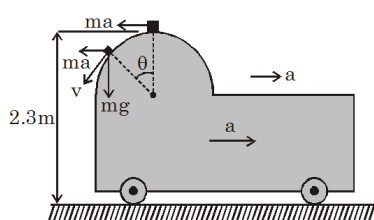
SECTION-I	Q.	1	2	3	4	5	6
	A.	A,B,D	A,B,C	A,B,D	A,B,C	A,C,D	B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	7.00	2.00	2.00	9.00	3.00	16.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	6	4	6	4	8	4

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,B,C	A,D	C,D	A,B,C,D	A,B,D	B,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	630.00	420.00	0.50	3.00	5.00	0.29
SECTION-III	Q.	1	2	3	4	5	6
	A.	4	2	7	6	4	3

HINT – SHEET
PART-1 : PHYSICS
SECTION-I

1. Ans (A,D)



$$maR \sin \theta + mgR (1 - \cos \theta) = \frac{1}{2}mv^2 \dots\dots(i)$$

$$mg \cos \theta - ma \sin \theta = \frac{mv^2}{R} \dots\dots(ii)$$

$$\therefore \theta = 37^\circ$$

$$2.3 - R (1 - \cos \theta) = v \sin \theta t + \frac{1}{2}gt^2$$

$$\therefore t = 0.51s \dots\dots(iii)$$

2. Ans (A,C)

First find the mass defect using the equation for mass defect.

$$\Delta m = Z (\text{atomic mass of H}) + Nm_n - \text{atomic mass}$$

To determine the number of oxygen-16 nuclei that must be formed to produce a total mass defect equal to the mass of a Caraphractus cinctus egg, calculate the ratio of the mass of one egg to the mass defect.

$$\text{the number of } {}^{16}_8\text{O nuclei formed} = n = \frac{m_{\text{egg}}}{\Delta m}$$

Finally, to find the total binding energy of all ${}^{16}_8\text{O}$ nuclei, use the equation for the binding energy of a nucleus and multiply it by n

$$\text{total } E_{\text{bind}} = nE_{\text{bind}} = n\Delta mc^2$$

$$\text{total } E_{\text{bind}} = n\Delta m (931.50 \text{ MeV/u})$$

$$\Delta m = 8(1.007825 \text{ u}) + 8(1.008665 \text{ u}) - 15.994915 \text{ u}$$

$$\Delta m = 8.062600 \text{ u} + 8.069320 \text{ u} - 15.994915 \text{ u}$$

$$\Delta m = 0.137005 \text{ u}$$

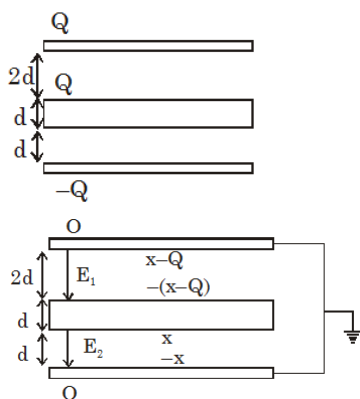
$$n = \frac{(2.0 \times 10^{-10} \text{ kg})}{(0.137005 \text{ u})} \left(\frac{1 \text{ u}}{1.66 \times 10^{-27} \text{ kg}} \right)$$

$$= 8.8 \times 10^{17} \text{ nuclei}$$

$$\text{total } E_{\text{bind}} = (8.8 \times 10^{17})(0.137005 \text{ u})(931.50 \text{ MeV/u})$$

$$\text{total } E_{\text{bind}} = 1.1 \times 10^{20} \text{ MeV}$$

3. Ans (A,C)



$$E_1 = \frac{x - Q}{\epsilon_0 A}$$

$$E_2 = \frac{x}{\epsilon_0 A}$$

$$\frac{x - Q}{\epsilon_0 A} 2d + \frac{x}{\epsilon_0 A} \cdot d = 0$$

$$\therefore x = \frac{2Q}{3}$$

4. Ans (B,D)

$$\epsilon = \frac{d\Phi}{dt} = -A \frac{dB}{dt} = -\left(d \cdot \frac{d}{2}\right) \frac{d(kt)}{dt} = -\frac{kd^2}{2}$$

is induced in each half-loop. For the duration that the B-field acts, E is the potential across each capacitor. Hence, each capacitor C_1 and C_2 acquires a charge given by the general formula $Q = CE$, or $Q_1 = -C_1 E$ and $Q_2 = C_2 E$. The different signs on the Q's are the result of the current in the opposite directions to ensure that Lenz's law is obeyed.

The central wire is now cut and the B-field held constant. With the B-field constant, there is no changing flux through the wire loop connecting the two capacitors, and E disappears. However, charges Q_1 and Q_2 must redistribute to bring both capacitors, now in parallel, to the same potential V. Treating the parallel capacitors as one, with capacitance $C = C_1 + C_2$ and a total charge

$$Q = Q_1 + Q_2 = \epsilon(C_2 - C_1) = \frac{kd^2}{2}(C_2 - C_1)$$

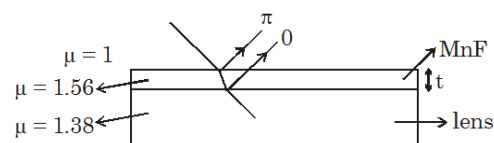
$$\text{we find } V = \frac{Q}{C} = \frac{(C_2 - C_1) kd^2}{(C_1 + C_2) 2}$$

Therefore, the resultant charges are :

$$Q_1 = C_1 V = \frac{(C_2 - C_1) C_1 kd^2}{(C_1 + C_2) 2}$$

$$\text{and } Q_2 = C_2 V = \frac{(C_2 - C_1) C_2 kd^2}{(C_1 + C_2) 2}$$

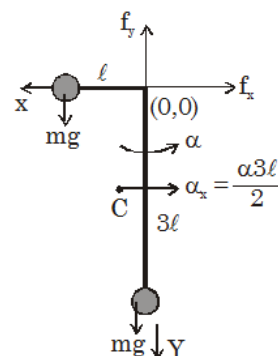
5. Ans (B,C,D)



For minima : $2ut = n\lambda$

6.

Ans (A,D)



$$Y_c = \frac{m \cdot 0 + m \cdot 3l}{2m} = \frac{3l}{2}$$

$$mg\ell = [m\ell^2 + m(3\ell)^2] \alpha$$

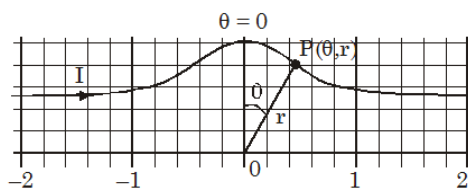
$$\Rightarrow \alpha = \frac{g}{10\ell}$$

$$f_x = 2m \left(\frac{\alpha 3\ell}{2} \right) = \frac{3mg}{10}$$

$$2mg - f_y = 2m \frac{\alpha \ell}{2} \Rightarrow f_y = \frac{19mg}{10}$$

SECTION-II

1. Ans (2.66 to 2.67)

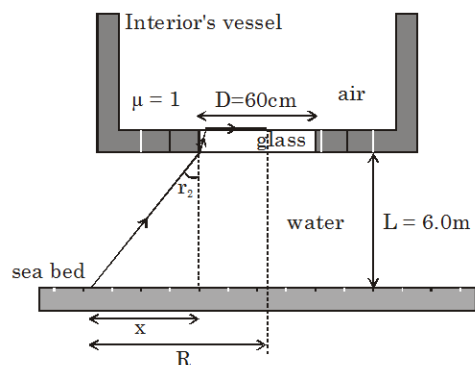


$$dB = \frac{\mu_0 i}{4\pi r^2} r d\theta$$

$$B = \frac{\mu_0 i}{4\pi} \int_{-\pi/2}^{\pi/2} \cos \theta (1 + \sin^2 \theta) d\theta$$

$$= \frac{2\mu_0 i}{3\pi}$$

2. Ans (157.00 to 160.33)



$$\frac{4}{3} \sin r_2 = 1.55 \sin r_1 = 1 \sin 90^\circ$$

$$\Rightarrow \sin r_2 = \frac{3}{4} \Rightarrow \tan r_2 = \frac{3}{\sqrt{7}}$$

$$\tan r_2 = \frac{x}{6} \Rightarrow x = \frac{18}{\sqrt{7}} \text{ m}$$

$$R = 0.3 + \frac{8}{\sqrt{7}}$$

$$\therefore A = \pi R^2$$

3. Ans (4.49 to 4.50)

$$\phi = NBA$$

$$\langle \epsilon \rangle = \frac{\phi_i - \phi_f}{t}$$

$$9.6 \times 10^{-3} = \frac{200 \times (0.68 - 0.24) \pi \times (0.025)}{4t}$$

$$\therefore t = 4.5 \text{ s}$$

4. Ans (112.50)

I will call h the height above the ground at which the projectile explodes and v_{jy} ($j = 1, 2, 3$) the y -component of the velocity at the moment of the explosion for the three fragments. I will label with 1 the fragment which lands after t seconds.

Since the landing time depends only on the y -component of the velocity at the moment of the explosion, we have:

$$v_{3y} = v_{2y}$$

We then need only the equations for fragments 1 and 2, stating that at landing time the y -coordinate will be equal to 0:

$$h + v_{1y}t - \frac{1}{2}gt^2 = 0$$

$$h + v_{2y}(2t) - \frac{1}{2}g(2t)^2 = 0$$

Since the masses of the three fragments are equal, and the y -component of the projectile momentum at the moment of the explosion is equal to 0 (it explodes at the top of the trajectory), the momentum conservation law reads:

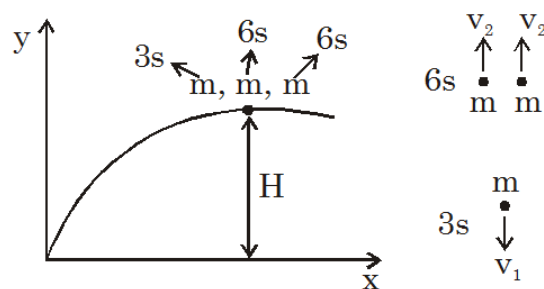
$$v_{1y} + v_{2y} + v_{3y} = 0.$$

The solution of the above equations is then:

$$v_{1y} = -\frac{3}{4}gt$$

$$v_{2y} = v_{3y} = \frac{3}{8}gt$$

The height above the ground at which the projectile explodes is then:



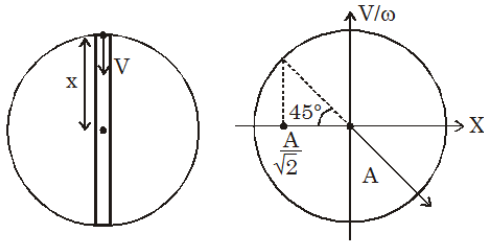
$$2mv_2 = mv_1 \dots (i)$$

$$H = v_1(3) + 5(3)^2$$

$$H = -v_2(6) + 5(6)^2$$

$$\therefore H = 112.5$$

5. Ans (19.77 to 19.80)



$$T = 2\pi\sqrt{\frac{R^3}{GM}} = 15 \text{ min} \quad \& \quad V = \sqrt{\frac{GM}{R}}$$

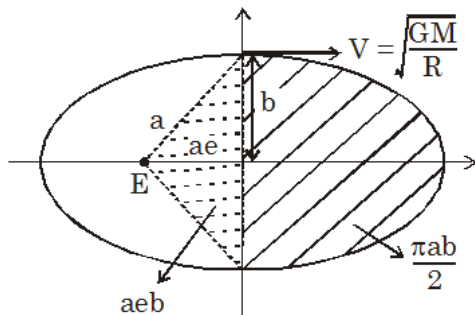
Inside earth, the rock will perform SHM

$$V = \omega\sqrt{A^2 - x^2} \Rightarrow \sqrt{\frac{GM}{R}} = \sqrt{\frac{GM}{R^3}} (A^2 - R^2)$$

$$\therefore A = \sqrt{2}R$$

$$\therefore \text{Time spent inside earth} = \frac{T}{8} \times 4$$

Outside earth



$$\pi ab \rightarrow T$$

$$\frac{\pi ab}{2} + eab \rightarrow \frac{T}{\pi} \left(\frac{\pi}{2} + e \right)$$

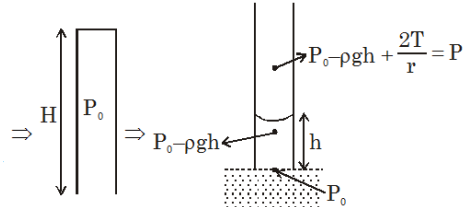
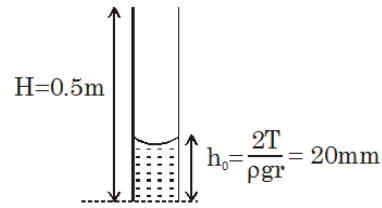
$$\Rightarrow T' = T \left(\frac{1}{2} + \frac{e}{\pi} \right)$$

For $e = 1, b = 0$

$$\therefore T' = T \left(\frac{1}{2} + \frac{1}{\pi} \right)$$

$$\therefore \text{Total time} = T \left(1 + \frac{1}{\pi} \right) = 19.77$$

6. Ans (0.94 to 0.95)

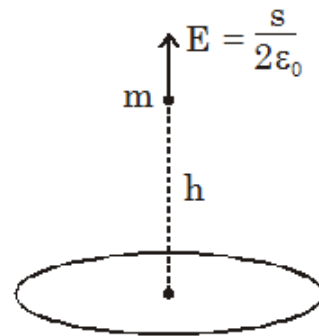


$$P_0 H = \left(P_0 - \rho gh + \frac{2T}{r} \right) \cdot (H - h)$$

Now solve :

SECTION-III

1. Ans (5)



$$mg = \frac{\sigma Q}{2\epsilon_0}$$

$$\Rightarrow mg - \frac{\sigma}{2\epsilon_0} \left[\frac{h}{\sqrt{R^2 + h^2}} \right] = ma$$

$$\Rightarrow \frac{\sigma Q}{2\epsilon_0} \left[1 - \left(1 + \frac{h^2}{R^2} \right)^{-\frac{1}{2}} \right] = ma$$

$$\therefore a = 5 \mu\text{m/s}^2$$

2. Ans (4)

$$C = C_V + \frac{R}{1-k} \Rightarrow 2R = \frac{3R}{2} + \frac{R}{1-k}$$

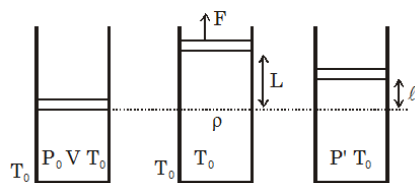
$$\therefore k = -1$$

$$\Rightarrow PV^{-1} = \text{constant}$$

$$\Rightarrow \frac{nRT}{V} \cdot \frac{1}{V} = \text{constant} \therefore T \propto V^2$$

$$\Rightarrow \frac{T^2}{T_1} = \left(\frac{2V}{V} \right)^2 = 4$$

3. Ans (2)



$$P_0 V = nRT_0 \text{(i)}$$

$$P_0 = \frac{mg}{A} \text{(ii)}$$

$$F + pA = mg \text{(iii)}$$

$$P (V + AL) = nRT_0 \text{(iv)}$$

$$P' (V + A\ell) = nRT \text{(v)}$$

$$P' = \frac{mg}{A} = P_0 \text{(vi)}$$

$$\text{From (i), (v) \& (vi): } \frac{nRT_0}{V} (V + A\ell) = nRT$$

$$\Rightarrow \frac{T}{T_0} = \frac{V + A\ell}{V} \text{ ... (vii)}$$

$$\text{Also, } T_0(V + AL)^{2/3} = T(V + A\ell)^{2/3} \text{ ... (viii)}$$

$$\text{From (vii) (viii): } \frac{V + A\ell}{V} = \left(\frac{V + AL}{V + A\ell} \right)^{2/3}$$

$$\Rightarrow (V + A\ell)^{5/3} = V(V + AL)^{2/3}$$

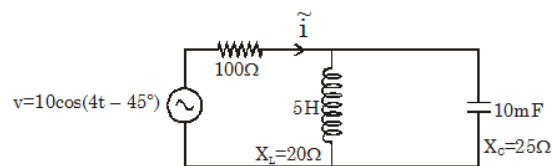
$$\Rightarrow V + A\ell = V^{3/5} (V + AL)^{2/5}$$

$$\Rightarrow A\ell = V \left[\left(1 + \frac{AL}{V} \right)^{2/5} - 1 \right]$$

$$\simeq V \left[1 + \frac{2AL}{5} - 1 \right] = \frac{2AL}{5}$$

$$\therefore \ell = \frac{2L}{5}$$

4. Ans (5)



$$\tilde{Z} = R + \frac{\tilde{X}_L \tilde{X}_C}{\tilde{X}_L + \tilde{X}_C}$$

$$= 100 + \frac{20j \times 25(-j)}{20j - 25j} = 100(1 + j)$$

$$\tilde{i} = \frac{\tilde{\varepsilon}}{\tilde{Z}} \therefore \tilde{v}_C = \tilde{i} (\tilde{Z} - R) = \frac{10e^{i(4t - \frac{\pi}{4})}}{100(1 + j)} \cdot 100j$$

$$= \frac{10}{2} e^{i(4t - \frac{\pi}{4})} (1 + j) = 5\sqrt{2}e^{j4t}$$

$$\text{at } t = \frac{\pi}{16} \text{ s ; } v_C = 5 \text{ V}$$

5. Ans (5)

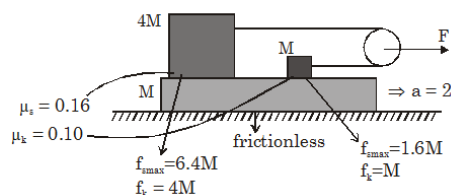
$$8 \text{ VSD} = 7 \text{ MSD} = 7 \text{ mm}$$

$$\therefore \perp \text{VSD} = \frac{7}{8} \text{ mm}$$

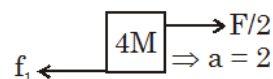
$$\therefore \text{LC} = \perp \text{MSD} - \perp \text{VSD} = \frac{1}{8} \text{ mm}$$

$$\therefore L = 4 + 5 \times \text{LC} = 4 + \frac{5}{8}$$

6. Ans (8)



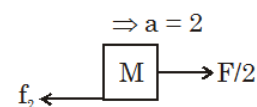
Case 1 : All move together $\Rightarrow F = (4M + M + M)a = 12M$



$$\frac{F}{2} - f_1 = 4M \times 2$$

$$\Rightarrow f_1 = \frac{F}{2} - 8M \leq 6.4M$$

$$\Rightarrow F \leq 28.8M$$



$$\frac{F}{2} - f_2 = M(2)$$

$$\Rightarrow f_2 = \frac{F}{2} - 2M \leq 1.6M$$

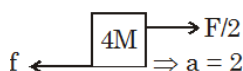
$$\Rightarrow F \leq 7.2M$$

Condition is not satisfied for block of mass M

$\therefore$ they cannot move together

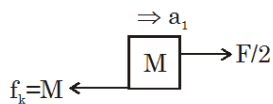
Case 2 :

'M' moves relative to plank



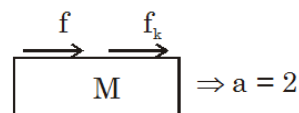
$$\frac{F}{2} - f = 4M(2)$$

$$\Rightarrow F = 18M$$



$$\Rightarrow \frac{F}{2} - M = Ma$$

$$\therefore a_1 = 8$$



$$f + f_k = M(2)$$

$$\therefore f = M$$

PART-2 : CHEMISTRY

SECTION-I

1. Ans (A,B,D)

$$V = 3 \text{ L}, 45.4 \text{ V}$$

$$V.S = M \times 11.35$$

$$45.4 = M \times 11.35 \Rightarrow M = 4 \text{ mole/L}$$

$$(A) \text{H}_2\text{O}_2 = 4 \times 3 = 12 \text{ mole}$$

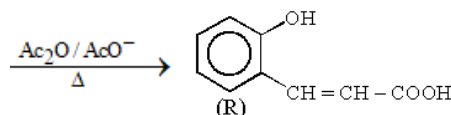
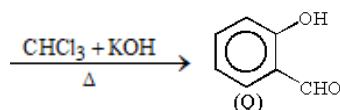
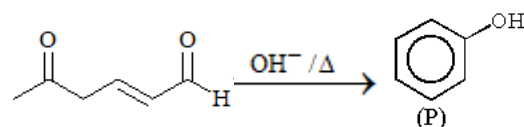
$$(B) V = 200 \times 60 \times 2.8 \text{ ml} = 33600 \text{ ml} = 33.6 \text{ L}$$

$$(C) \text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{O} + \frac{1}{2} \text{O}_2, n_{\text{O}_2 \text{ used}} = \frac{33.6}{22.4} = \frac{3}{2} = 1$$

$$M = \frac{9}{3} = 3$$

$$V.S = 3 \times 11.35 = 34.05$$

5. Ans (A,C,D)



SECTION-II

1. Ans (7.00)

$$\Delta U = q + W$$

$$q = -W = P_A V_A \ln \frac{V_B}{V_A} = 100 \times 1 \ln \frac{2}{1}$$

$$= 100 \times 0.7 = 70 \text{ bar L}$$

$$70 \times 0.1 \text{ kJ} = 7 \text{ kJ}$$

2. Ans (2.00)

$$\lambda_1 = 10 \text{ cm}; P_1 = 1 \text{ bar}$$

$$\lambda_2 = ?; P_2 = 5 \text{ bar}, T = \text{constant.}$$

$$\lambda \propto \frac{1}{P} \Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{P_2}{P_1}$$

$$\text{or, } \frac{10}{\lambda_2} = \frac{5}{1} \Rightarrow \lambda_2 = 2 \text{ cm}$$

3. Ans (2.00)

$$\frac{6+0}{3} = 2.00$$

SECTION-III

1. **Ans (6)**

At node,

$$\Psi^2(r) = 0$$

$$\Rightarrow \Psi(r) = 0$$

$$\ell = 0$$

$$n - \ell - 1 = 3 \Rightarrow n = 4$$

$$\Rightarrow (1 - \sigma)(\sigma^2 - 8\sigma + 12) = 0$$

$$\text{when } 1 - \sigma = 0 \Rightarrow \sigma = 1$$

$$\text{or, } \frac{2r}{4a_0} = 1 \Rightarrow r = 2a_0$$

$$\text{when } \sigma^2 - 8\sigma + 12 = 0$$

$$\Rightarrow \sigma = \frac{8 \pm \sqrt{64 - 48}}{2}$$

$$\text{or, } \sigma = \frac{8 \pm 4}{2} = 6 \text{ and } 2$$

$$\text{when } \sigma = 6$$

$$\Rightarrow \frac{2r}{4a_0} = 6$$

$$\text{or, } \frac{2r}{4a_0} = 6 \Rightarrow r = 12a_0$$

$$\text{when } \sigma = 2$$

$$\Rightarrow \frac{2r}{na_0} = 2 \Rightarrow \frac{r}{4a_0} = 1 \Rightarrow r = 4a_0$$

$$\therefore \text{distance of nearest radial node} = 2a_0 = xa_0$$

$$\Rightarrow x = 2$$

$$\text{and distance of farthest radial node} = 12a_0 = ya_0$$

$$\Rightarrow y = 12$$

$$\therefore y/x = 12/2 = 6 \text{ ans.}$$

2. **Ans (4)**

$$\Lambda_{mHX} = \frac{\Lambda_{mHY}}{100}$$

$$[HX] = [HY] = 0.1 \text{ M} = C$$

$$\therefore \lambda_{x-}^0 = \lambda_{y-}^0$$

$$\Rightarrow \Lambda_{m,HX}^0 = \Lambda_{m,HY}^0$$

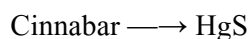
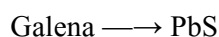
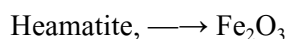
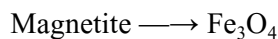
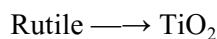
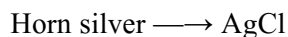
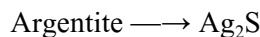
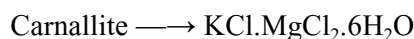
$$\frac{K_{aHX}}{K_{aHY}} = \frac{C\alpha_{HX}^2}{C\alpha_{HY}^2} = \left(\frac{\Lambda_{mHX}}{\Lambda_{mHY}} \right)^2 = \left(\frac{1}{100} \right)^2 = 10^{-4}$$

$$\text{or, } \log K_{aHX} - \log K_{aHY} = -4$$

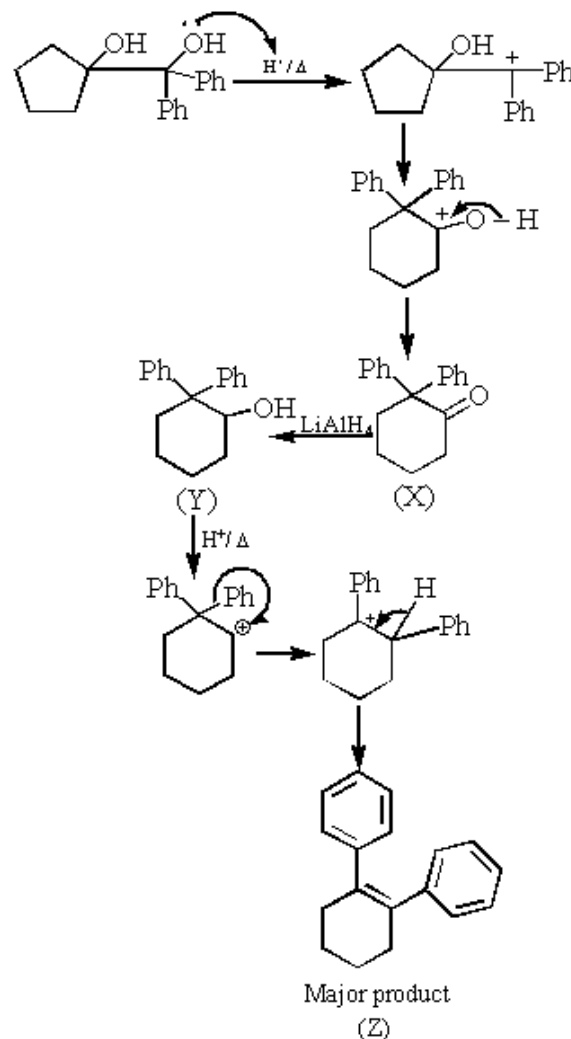
$$\text{or, } pK_{aHX} + pK_{aHY} = -4$$

$$\text{or, } pK_{aHX} - pK_{aHY} = 4$$

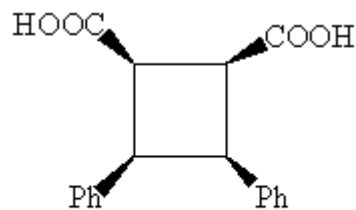
3. **Ans (6)**



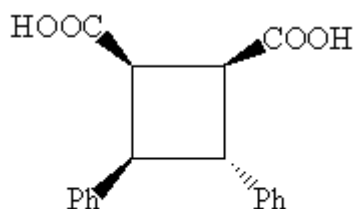
5. **Ans (8)**



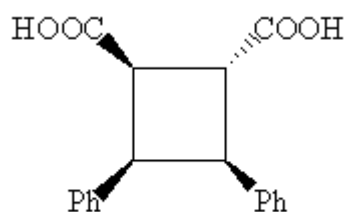
6. Ans (4)



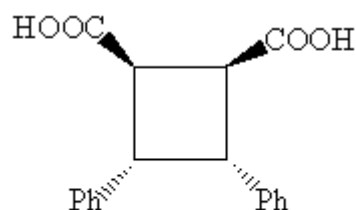
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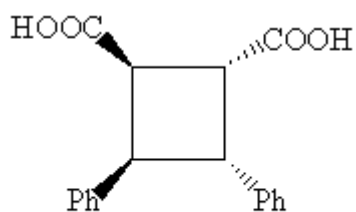
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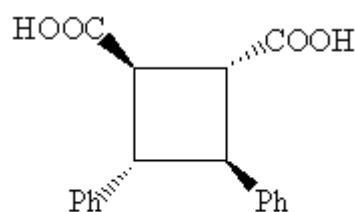
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Meso



dl



dl

PART-3 : MATHEMATICS

SECTION-I

1.

Ans (A,B,C)

$$|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$\Rightarrow |A| = \begin{vmatrix} a_{11} - a_{12} & a_{12} - a_{13} & a_{13} \\ a_{21} - a_{22} & a_{22} - a_{23} & a_{23} \\ a_{31} - a_{32} & a_{32} - a_{33} & a_{33} \end{vmatrix}$$

Let $a_{11} = (2n + 1)^2$, $a_{12} = (2m + 1)^2$ and so on.

$$\Rightarrow a_{11} - a_{12} = (2n + 1)^2 - (2m + 1)^2$$

$$= 4(n^2 - m^2 + n - m)$$

$$= 4(n - m)(n - m + 1)$$

$$= 4k(k + 1)$$

$$= 8\lambda$$

$\Rightarrow$ two column of $|A|$ are multiple of 8

$\Rightarrow |A|$ is divisible by 64

Now, $\because |C| = |A|^2 \Rightarrow |C|$ is also divisible by 64.

$$\because a_{ij} \in [1, 1000]$$

$$\Rightarrow a_{ij} \in \{1^2, 3^2, 5^2, \dots, 31^2\}$$

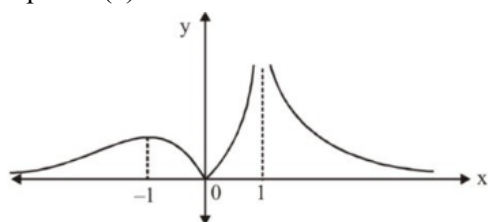
$$\Rightarrow \text{No. of such matrices} = {}^{16}C_9 \cdot 9!$$

2. Ans (A,D)

$$f(x) = \begin{cases} \frac{x}{(x-1)^2} & ; x > 0, x \neq 1 \\ \frac{-x}{(x-1)^2} & ; x < 0 \end{cases}$$

$$f'(x) = \begin{cases} \frac{-(x+1)}{(x-1)^3} & ; x > 0, x \neq 1 \\ \frac{x+1}{(x-1)^3} & ; x < 0 \end{cases}$$

Graph of $f(x)$ is



From the graph it is clear that $f(-1) = 0$ and Rolle's theorem is applicable in $[a, b]$ where $a < -1$ and $-1 < b < 0$.

3. Ans (C,D)

$$\begin{aligned} |P| &= 32 \Rightarrow |Q| |P|^2 = |Q|^3 |Q|^3 \\ \Rightarrow |Q|^5 &= (32)^2 \Rightarrow |Q| = 4 \Rightarrow QP^2 = 4Q^3 \\ \Rightarrow Q^{-1}QP^2 &= 4Q^{-1}Q^3 \Rightarrow P^2 - 4Q^2 = 0 \\ \Rightarrow (P + 2Q)(P - 2Q) &= 0 \quad (\because PQ^2 = Q^2P) \dots (1) \end{aligned}$$

Let $|P + 2Q| \neq 0$

$$\Rightarrow (P + 2Q)^{-1}(P + 2Q)(P - 2Q) = 0$$

$$\Rightarrow P - 2Q = 0 \text{ (which contradicts)}$$

$$\Rightarrow |P + 2Q| = 0$$

from (1) $(P + 2Q)R = 0$ (where $R = P - 2Q \neq 0$)

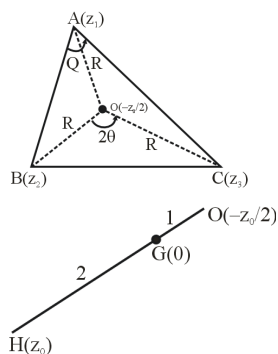
$\Rightarrow$ There exists non null matrices R such that

$$PR + 2QR = 0$$

4. Ans (A,B,C,D)

$$z_1^3 + z_2^3 + z_3^3 - 3z_1z_2z_3 = 0 \Rightarrow z_1 + z_2 + z_3 = 0$$

$$\underbrace{z_1^2 + z_2^2 + z_3^2 - z_1z_2 - z_2z_3 - z_3z_1 = 0}_{\text{(rejected } \because \text{Non-equilateral)}}$$



(A) $\because$ distance between vertices and orthocentre are given by

$$2R\cos A, 2R\cos B, 2R\cos C$$

$$\text{and } \cos A + \cos B + \cos C < \frac{3}{2}$$

$$2R\cos A + 2R\cos B + 2R\cos C < 3R$$

$$\Rightarrow |z_1 - z_0| + |z_2 - z_0| + |z_3 - z_0| < 3 \left| z_1 + \frac{z_0}{2} \right|$$

$$(B) \text{ amp } \left(\frac{z_3 + \frac{z_0}{2}}{z_2 + \frac{z_0}{2}} \right) = 2 \text{ amp } \left(\frac{z_3 - z_1}{z_2 - z_1} \right)$$

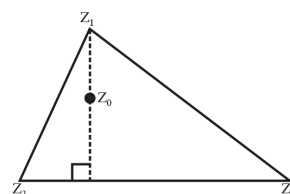
(C) circumcentre is

$$\frac{-z_0}{2} \text{ and } R = \left| z_1 + \frac{z_0}{2} \right|$$

$$\Rightarrow \text{equation of circumcircle is } \left| z + \frac{z_0}{2} \right| = R$$

$$\Rightarrow \left| z + \frac{z_0}{2} \right| = \left| z_1 + \frac{z_0}{2} \right|$$

(D) $\because z_1 - z_0$ is perpendicular to $z_3 - z_2$



$$\Rightarrow \frac{z_1 - z_0}{z_3 - z_2} \text{ is purely imaginary}$$

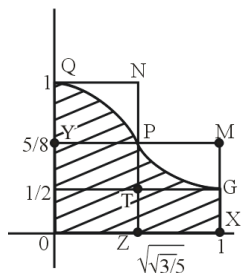
$$\Rightarrow \frac{\bar{z}_1 - \bar{z}_0}{\bar{z}_3 - \bar{z}_2} = -\frac{z_1 - z_0}{z_3 - z_2}$$

5. Ans (A,B,D)

$$y \leq \frac{1}{x^4 + 1}, x \leq 1, y \geq 0, \text{ so region R is shaded in}$$

given figure.

$$\frac{dy}{dx} = \frac{-4x^3}{(x^4 + 1)^2} \downarrow$$



$$\frac{d^2y}{dx^2} = 0 \text{ at } x = \left(\frac{3}{5}\right)^{\frac{1}{4}} \text{ (inflection)}$$

(A) $S_R \geq$ sum of area of rectangles OYPZ and TGXZ

$$\geq \frac{5}{8} \sqrt{\frac{3}{5}} + \frac{1}{2} \left(1 - \sqrt{\frac{3}{5}}\right)$$

$$\geq \frac{1}{8} \sqrt{\frac{3}{5}} + \frac{1}{2}$$

(B) $S_R \leq$ sum of area of rectangles OQNZ and

PMXZ

$$\leq \sqrt{\frac{3}{5}} + \frac{5}{8} \left(1 - \sqrt{\frac{3}{5}}\right)$$

$$\leq \frac{5}{8} + \frac{3}{8} \sqrt{\frac{3}{5}}$$

(C) $S_R \geq$ area of trapezium OQPZ

$$\geq \frac{1}{2} \left(1 + \frac{5}{8}\right) \cdot \sqrt{\frac{3}{5}}$$

$$\geq \frac{13}{16} \sqrt{\frac{3}{5}}$$

$$(D) \because S_R = \int_0^1 \frac{dx}{x^4 + 1}$$

$$\geq \int_0^1 \frac{dx}{x^2 + 1} \geq \frac{\pi}{4}$$

6. Ans (B,D)

$$f(x) = e^{x-1} + 2 - |2x - 1| + |e^{x-1} - (2 - |2x - 1|)|$$

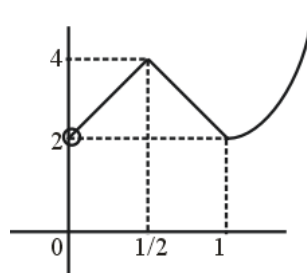
$$= a + b + |a - b|$$

$$= 2 \max\{a, b\}$$

$$\Rightarrow f(x) = 2 \max\{a, b\}$$

$$\Rightarrow f(x) = 2 \max\{e^{x-1}, 2 - |2x - 1|\}$$

Range of $f(x)$ $[2, \infty)$



$$x_0 > \frac{1}{2} \Rightarrow x_0 = 1$$

RHD at $x = 1$

$$= \left[\frac{d}{dx} (2e^{x-1}) \right]_{x=1} = 2$$

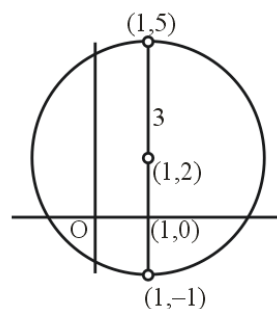
PART-3 : MATHEMATICS

SECTION-II

1. Ans (630.00)

$$-1 < a - 1 < 5$$

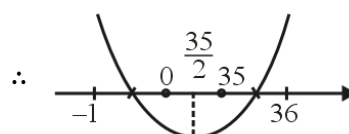
$$\Rightarrow 0 < a < 6$$



Now,

$$f(x) = x^2 - 35x - a^2 < 0$$

$$\because D = 35^2 + 4a^2 > 0$$



$$f(0) < 0, f(35) = -a^2 < 0, f(36) = 36 - a^2 > 0$$

$\Rightarrow$ all integral values of x are

$$0, 1, 2, 3, \dots, 35$$

$\Rightarrow$ sum of all values = 630

2.

Ans (420.00)

$$I = \int_0^{\pi} \frac{\cos 2nx - \cos 6nx}{\sin^2 x} dx$$

$$= \int_0^{\pi} \frac{(1 - 2\sin^2 nx) - (1 - 2\sin^2 3nx)}{\sin^2 x} dx$$

$$= 2 \left[\int_0^{\pi} \frac{\sin^2 3nx}{\sin^2 x} dx - \int_0^{\pi} \frac{\sin^2 nx}{\sin^2 x} dx \right]$$

$$\text{Let } I_n = \int_0^{\pi} \frac{\sin^2 nx}{\sin^2 x} dx$$

$$\Rightarrow I_n - I_{n-1} = \int_0^{\pi} \frac{\sin^2 (nx) - \sin^2 (n-1)x}{\sin^2 x} dx$$

$$= \int_0^{\pi} \frac{\sin(2n-1)x}{\sin x} dx$$

$$= \pi$$

$$\Rightarrow I_n - I_{n-1} = \pi$$

$$I_1, I_2, I_3, \dots, I_n \text{ in A.P. with } I_1 = \pi, d = \pi$$

$$\pi, 2\pi, 3\pi, \dots, n\pi \text{ are value of } I_1, I_2, I_3, \dots, I_n$$

$$\Rightarrow I_n = n\pi \text{ and } I_{3n} = 3n\pi$$

$$\text{Now, } I = 2(I_{3n} - I_n)$$

$$= 2.2n\pi$$

$$I = 4n\pi$$

$$\Rightarrow \sum_{n=1}^{20} I = 4\pi \sum_{n=1}^{20} n$$

$$= 4\pi(210)$$

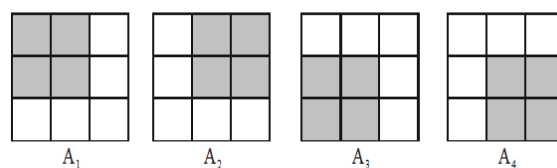
$$\Rightarrow \frac{1}{2\pi} \sum_{n=1}^{20} I = 420$$

3. Ans (0.50)

We will do this by complimentary counting. Let x be the number of ways in which we have atleast one square of size, 2×2 which Red, obviously for Green also number of ways too is x .

$$\text{Hence our answer} = 2^9 - 2x$$

No consider these situations



$$x = n(A_1 \cup A_2 \cup A_3 \cup A_4)$$

$$= 4.2^5 - \{2^3 + 2^3 + 2^2 + 2^2 + 2^2 + 2^3 + 2^3\} + 4.2^1 - 1$$

$$= 128 - 40 + 8 - 1 = 136 - 41 = 95$$

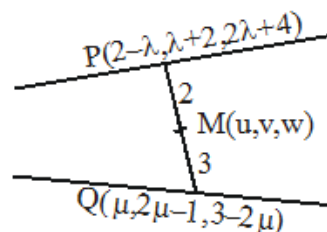
$$\text{Hence required answer} = 512 - 190 = 322$$

$$\Rightarrow p = \frac{322}{512}$$

4. Ans (3.00)

$$L_1 : \frac{x-2}{-1} = \frac{y-2}{1} = \frac{z-4}{2} \text{ and}$$

$$L_2 : \frac{x-0}{1} = \frac{y+1}{2} = \frac{z-3}{-2}$$



Using section formula and eliminating

$$\lambda, \mu \text{ we get } w + 2u = 6$$

$$M(u, v, w) \text{ moves on}$$

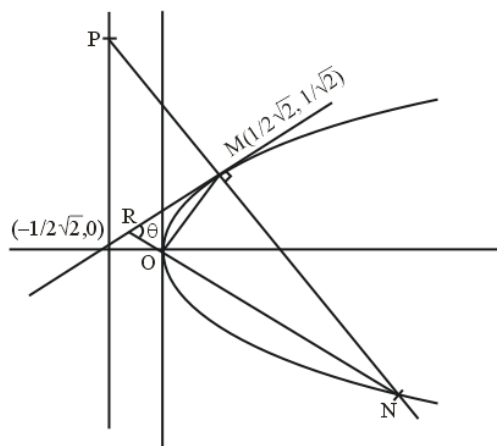
$$2x + z = 6$$

5. Ans (5.00)

$$y^2 = \sqrt{2}x \Rightarrow a = \frac{1}{2\sqrt{2}}$$

its directrix $x = -\frac{1}{2\sqrt{2}}$ intersects

the line $2\sqrt{2}x - \sqrt{2}y + 3 = 0$ at $P \left(-\frac{1}{2\sqrt{2}}, \sqrt{2} \right)$



equation of normal is $y = mx - 2am - am^3$

passing through $\left(-\frac{1}{2\sqrt{2}}, \sqrt{2} \right)$

$$\Rightarrow \sqrt{2} = \frac{-1}{2\sqrt{2}}m - 2\frac{1}{2\sqrt{2}}m - \frac{1}{2\sqrt{2}}m^3$$

$$m^3 + 3m + 4 = 0$$

$$\Rightarrow m = -1$$

foot of normal is $M(am^2, -2am) \equiv M \left(\frac{1}{2\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$

$$\therefore t_1 = 1 \Rightarrow t_2 = -t_1 - \frac{2}{t_1} = -3$$

$\Rightarrow$ other end of normal is $N \left(\frac{9}{2\sqrt{2}}, -\frac{3}{\sqrt{2}} \right)$

$$\text{slope of RN} = m_{ON} = -\frac{2}{3}$$

slope of RM = 1

$$\Rightarrow \tan \theta = \left| \frac{1 + \frac{2}{3}}{1 - \frac{2}{3}} \right| = 5$$

6. Ans (0.29)

$$\frac{dy}{dx} - y = -e^x \tan^2 x \text{ IF} = e^{-x}$$

$$\Rightarrow y \cdot e^{-x} = - \int e^{-x} \cdot e^x \tan^2 x \, dx$$

$$= \int (1 - \sec^2 x) \, dx$$

$$= x - \tan x + c$$

$$\text{as } x \rightarrow 0^+, y \rightarrow 0 \Rightarrow c = 0$$

$$\Rightarrow y = e^x (x - \tan x)$$

Now, $y = e^x(x^2 - \tan x)$ and $y = e^x(x - \tan x)$

intersects $\Rightarrow x = x^2 \Rightarrow x = 0, 1 \Rightarrow$ area bounded

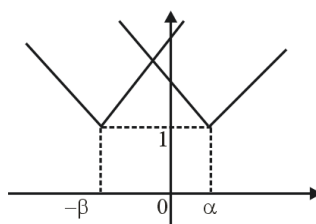
$$A = \left| \int_0^1 (e^x (x - \tan x) - e^x (x^2 - \tan x)) \, dx \right|$$

$$= \left| \int_0^1 e^x (x - x^2) \, dx \right| = 3 - e$$

SECTION-III

1. Ans (4)

$$f(x) = \frac{1}{|x - \alpha| + 1} + \frac{1}{|x + \beta| + 1}; 0 < \alpha < \beta$$



clearly $|x - \alpha| + 1$ and $|x + \beta| + 1$ are $\uparrow$ in (α, ∞)

hence $f \downarrow$ for $x > \alpha$

and $|x - \alpha| + 1$ and $|x + \beta| + 1$ are $\downarrow$ in $(-\infty, -\beta)$

hence f is $\uparrow$ for $x < -\beta$

Now,

$$f(x) = \frac{1}{\alpha + 1 - x} + \frac{1}{x + \beta + 1} \text{ in } (-\beta, \alpha)$$

$$= \frac{(x + \beta + 1)^2 - (x - (\alpha + 1))^2}{(\alpha + 1 - x)^2 (x + \beta + 1)^2}$$

$$= (2x + \beta - \alpha) \cdot (\alpha + \beta + 2)$$

$$\begin{array}{c} \downarrow \quad \quad \quad \uparrow \\ x_0 = \frac{\alpha - \beta}{2} \text{ (minima)} \end{array}$$

$$f(x_0) = \frac{2}{|\alpha - \beta - 2\alpha| + 2} + \frac{2}{|\alpha - \beta + 2\beta| + 2}$$

$$= \frac{4}{\alpha + \beta + 2}$$

2. Ans (2)

$$\frac{P(x)}{(2x-3)^2(3x-2)^2} = \frac{A}{2x-3} + \frac{B}{(2x-3)^2} + \frac{C}{3x-2} + \frac{D}{(3x-2)^2}$$

$$\int \frac{P(x)dx}{(2x-3)^2(3x-2)^2} = \frac{1}{5} \ln |f(x)| + C \text{ and}$$

$f(x)$ is rational function

$$B = D = 0$$

$$\text{and } P(x) = \lambda(2x-3)(3x-2)$$

$$P(1) = -\lambda = -1 \Rightarrow \lambda = 1$$

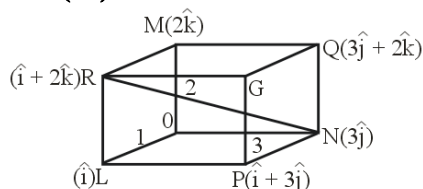
$$\int \frac{dx}{(2x-3)(3x-2)} = \int \left[\frac{2}{2x-3} - \frac{3}{3x-2} \right] \frac{dx}{5}$$

$$= \frac{1}{5} \ln \left| \frac{2x-3}{3x-2} \right| + C = \frac{1}{5} \ln \left| \frac{4x-6}{3x-2} \right| + C'$$

$$\therefore \lim_{x \rightarrow \infty} f(x) = \frac{4}{3} \Rightarrow f(x) = \frac{4x-6}{3x-2}$$

$$\Rightarrow |f(1)| = 2$$

3. Ans (7)



$$\text{equation of OL : } \vec{r} = 0 + \mu \hat{i}$$

$$\text{equation of RN : } \vec{r} = 3\hat{j} + \lambda (\hat{i} - 3\hat{j} + 2\hat{k})$$

(skew body diagonal)

$$\Rightarrow \ell = S.D = \left| (3\hat{j}) \cdot \hat{n} \right|, \vec{n} = (\hat{i} - 3\hat{j} + 2\hat{k}) \times \hat{i}$$

$$= \left| 3\hat{j} \cdot \frac{(2\hat{j} + 3\hat{k})}{\sqrt{13}} \right| = \frac{6}{\sqrt{13}}$$

$$\text{equation of ON : } \vec{r} = 0 + \mu (3\hat{j})$$

$$\text{equation of QL : } \vec{r} = \hat{i} + \lambda (\hat{i} - 3\hat{j} - 2\hat{k})$$

(skew body diagonal)

$$\Rightarrow n = S.D = \left| \hat{i} \cdot \hat{n} \right|, \vec{n} = (\hat{i} - 3\hat{j} - 2\hat{k}) \times \hat{j}$$

$$= \left| \hat{i} \cdot \frac{(-2\hat{i} - \hat{k})}{\sqrt{5}} \right| = \frac{2}{\sqrt{5}}$$

$$\text{equation of OM : } \vec{r} = 0 + \lambda (2\hat{k})$$

$$\text{equation RN : } \vec{r} = 3\hat{j} + \lambda (\hat{i} - 3\hat{j} + 2\hat{k})$$

(skew body diagonal)

$$m = S.D = \left| 3\hat{j} \cdot \hat{n} \right|, \vec{n} = (\hat{i} - 3\hat{j} + 2\hat{k}) \times \hat{k}$$

$$= \left| 3\hat{j} \cdot \frac{(-3\hat{i} - \hat{j})}{\sqrt{10}} \right| = \frac{3}{\sqrt{10}}$$

$$\Rightarrow \sqrt{13\ell^2 + 10m^2 + 5n^2} = 7$$

4. Ans (6)

Let

$$P = P((A \cap B) / (A \cup \bar{B})) = \frac{P(A \cap B)}{P(A \cup \bar{B})}$$

$$\therefore A \cap B \subset A \cap \bar{B}$$

$$\Rightarrow p = \frac{P(A \cap B)}{1 - (P(B) - P(A \cap B))}$$

$$= \frac{1/4}{1 - \frac{1}{3} + \frac{1}{4}}$$

$$= \frac{3}{11} \Rightarrow k = 6$$

5. Ans (4)

$$\frac{{}^7C_2}{(2^7 - 1)^f} = \frac{21}{(127)^4}$$

$$\Rightarrow (2^7 - 1)^f = (127)^4 \Rightarrow f = 4$$

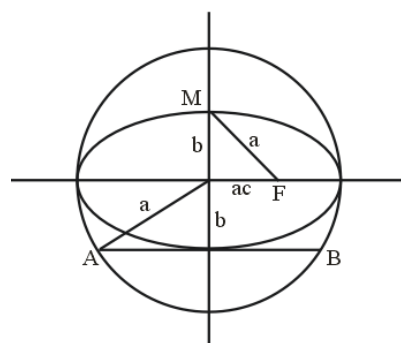
6. Ans (3)

$$E_i = \frac{x^2}{a_{i+1}^2} + \frac{y^2}{a_i^2} = 1$$

its auxiliary circle is

$$C_i \equiv x^2 + y^2 = a_{i+1}^2$$

$$e_i = \sqrt{1 - \left(\frac{a_i}{a_{i+1}} \right)^2} = \sqrt{1 - \frac{1}{4}} = \frac{\sqrt{3}}{2}$$



$$FM = a_{n+1} \text{ for } E_n \equiv \frac{x^2}{a_{n+1}^2} + \frac{y^2}{a_n^2}$$

$$L = AB = 2\sqrt{a^2 - b^2} = 2ac = 2a_{n+1} \frac{\sqrt{3}}{2}$$

$$\Rightarrow \frac{L}{FM} = \sqrt{3}$$